Answers and talk notes

Division Expressions with Fractions and Mixed Numbers

What your student is learning

Tonight we divide mixed numbers — and the whole trick is turning them into improper fractions BEFORE anything else.

Answers

Ask your student to explain before you look.

1. **B.** 32/5. 6 2/5 = 32/5, because 6 × 5 = 30 and 30 + 2 = 32 fifths.
2. **D.** 15/4. 3 3/4 = 15/4, because 3 × 4 = 12 and 12 + 3 = 15 fourths.
3. **A.** Write each mixed number as an improper fraction. Write each mixed number as an improper fraction first. Only then can you use Keep, Change, Flip.
4. **C.** 2. 7/2 ÷ 7/4 = 7/2 × 4/7 = 28/14 = 2.
5. **B.** 2/3. 1 1/2 = 3/2 and 2 1/4 = 9/4, so 3/2 × 4/9 = 12/18 = 2/3.

Talk about it

What are your two improper fractions, and how many pieces fit?

Home mission

Grab any string or ribbon and pretend it is 6 3/4 meters long, with 2 1/4 meters for each bow. Ask: about how many bows? Then count in quarters: how many quarters are in each length?

When they are stuck, ask

- Is every number a fraction yet?
- Which fraction gets flipped — the first or the second?
- How many groups of that size fit inside the total?

Try not to say

- Divide the whole numbers, then divide the fractions.
- Flip both fractions to be safe.
- Leave it as a mixed number, it is easier to read.